Agricultural VCU v2 PCB Test Report

1. Introduction

The Agricultural VCU v2 PCB is developed as the control unit platform for agricultural robots and machineries. Following fabrication and assembly, a structured test process was required to verify that the manufactured board could operate as intended and could support subsequent firmware and interface validation. This test report documents the verification activities carried out on the V2 PCB as the current stage of hardware evaluation.

The report provides a technical record of the device under test, the assembly condition, the test setup, and the procedures used to evaluate the board. It covers the essential preparation steps before interface validation, including device detection through the SWD programming path, the use of laboratory instruments, and the use of test firmware for functionality test. In addition to presenting the measurement results, the report also serves to identify observed issues during testing and to support future design refinement and debugging activities.

The focus of the present evaluation is testing the implemented interfaces. Each interface is tested independently using dedicated firmware so that any abnormal behaviour is easier to trace to a specific circuit block. The validated or investigated functions include MCU debugging access, GPIO, relay control, PWM generation, ADC, DAC, UART, and CAN communication, while the remaining unverified or incomplete tests are noted for follow-up work. By separating the tests in this manner, the report provides a clearer assessment of which subsystems are already operational and which still require further verification or proof-of-concept measurement.

2. Test Setup

2.1. Device Under Test (DUT)

Agricultural VCU v2 PCB

PCB Specifications

Specifications	PCB V2
Layer Count	4
Dimensions	176.53 mm x 184.15 mm
PCB Thickness	2 mm
Minimum track width	0.1 mm
Minimum track spacing	0.1 mm
Minimum hole diameter	0.2 mm

PCB Assembly Process

We have 10 PCBs with several components, as shown in the table below, soldered by the manufacturer, and 2 bonus bare PCBs from the manufacturer.

Component	Series Number	Note
Integrated circuit (IC)	M467HJHAN	Microcontroller (MCU) IC
Integrated circuit (IC)	RTL8201FI-VC-CG	Ethernet PHY IC
Crystal	TSX-3225 25.0000MF10P-C0	25 MHz Crystal

During initial testing, instability was observed on the 3V3 rail, so the standard assembly flow could not be continued directly on all boards. As a result, the bare PCBs were used

for additional investigation of the regulator behaviour and references for controlled re-assembly and testing.

It was found that removing the voltage regulator TPS79533 output capacitor (C3) makes the 3V3 rail stable around 3.3V. Once the cause of instability in the 3V3 rail is identified, the assembly strategy is adjusted to facilitate troubleshooting. The power-supply section was prioritised and verified first, followed by passive components added block by block, with partial power checks performed after each stage. The MCU and other critical active components were deferred until the required power rails, especially the 3V3 rail, were confirmed stable. This modified process is adopted to reduce debugging complexity and prevent further problems. All connectors are not placed during testing unless necessary.

Assembled PCB

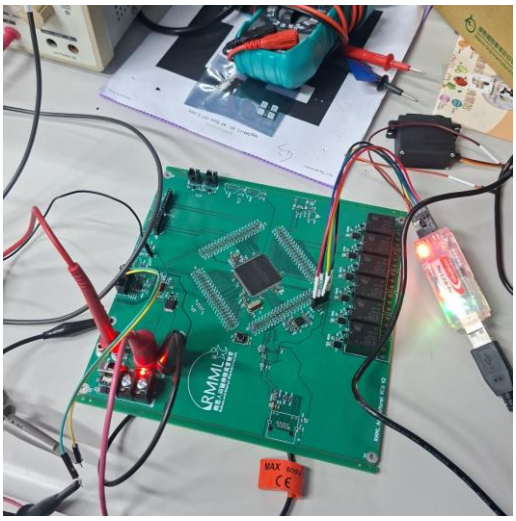
Top View	Bottom View
	

2.2. Test Equipment

The equipment used to check trace connectivity, measure voltage and signals, and test the firmware are shown in table below

Equipment	Note
PCB	Manufactured by EMBPCB
Multimeter	ProsKit MT-1220
Power supply	GWInstek GPS-3303 3CH
Oscilloscope	GWInstek GDS-1102-U
Breadboard	
Jump wire	

2.3. Test Setup



3. Power Test

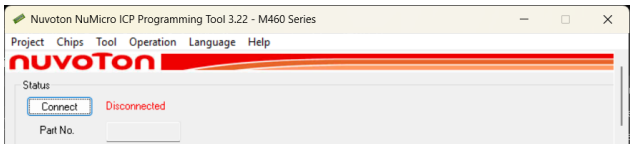
Due to limited stock of buck Module, the 12V and 5V rails are supplied independently from DC power supply. 3V3 is obtained from the voltage regulator TPS79533. If the measured 3V3 rail is not exactly 3.3V, check the voltage on the IC's feedback pin. The tolerable value of feedback voltage of the IC is 1.2V, which is stated on TPS79533 datasheet, and slight difference of ±0.01V is tolerable. The following data represents the quiescent power state, measured immediately after power-on without any external load or debugging activity.

Test Result:

V_{12V}	V_{5V}	V_{3V3}	V_{FB}
12V	5.03V	3.33V	1.19V

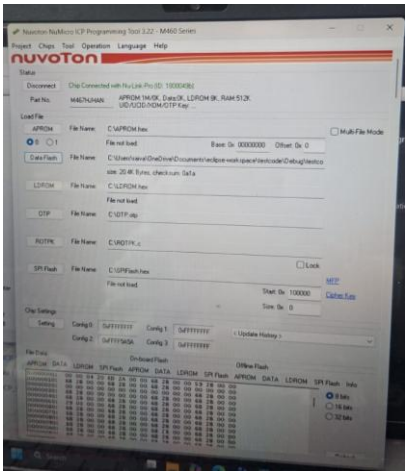
The current of each power rail during the quiescent power test is relatively small, as example, I_{5V} is measured around 0.09 A and I_{3V3} from output regulator is 265µA

4. Debug Test



Before conducting an interface test, the MCU must first be verified as detectable and programmable through the debugging interface. This verification was performed using the NuMicro ICP Programming Tool. When the connection is successful, the software displays the connected debugger information, the target MCU type, and the corresponding device identification data, confirming that the SWD programming path and the essential MCU support circuitry are functioning correctly.

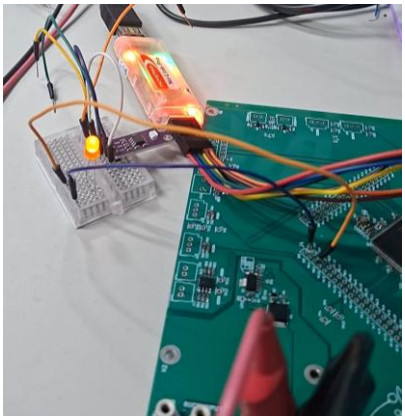
After successful device detection, a test firmware file in .hex format was loaded into the APROM memory region using the same programming tool. The program was then executed to confirm that the MCU could be programmed and debugged without communication failure. The result of this test indicates that the MCU on the PCB was successfully recognized by the programming tool and that the board satisfied the basic requirement for subsequent firmware download and debugging activities.



5. Interface Test

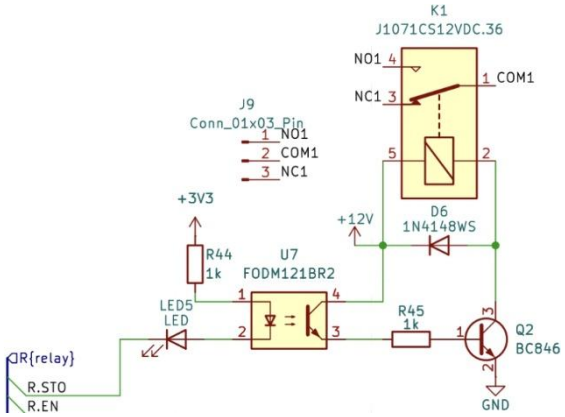
All interfaces are tested independently with separate firmware. If an issue is found during testing one of the interfaces, it would be easier to troubleshoot than testing them all simultaneously. The code used for testing each interface refers from Nuvoton's board service package (BSP) on Github.

5.1. GPIO



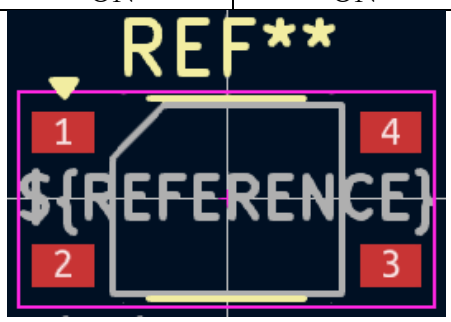
GPIO functionality is verified using a simple LED circuit on a breadboard, where several GPIO pins were selected randomly for testing. Each selected pin was configured as an output and successfully toggled the LED state according to the uploaded program. The test result confirmed that the GPIO interface is operating properly, and no abnormal behavior was observed during this stage.

5.2. Relay



Based on the schematic, the relay-driving circuit was initially expected to operate correctly through the optocoupler and transistor stages. However, initial testing showed that the actual relay behaviour did not match the intended operation. Further troubleshooting revealed that the pull-up voltage on the low-voltage side of the optocoupler input stage was insufficient for proper switching. A significant portion of the available voltage was dropped across the external indicator LED and the optocoupler's internal LED, leaving insufficient drive to activate the optocoupler's transistor effectively. Using too high voltage will keep the relay in switching condition, regardless of the logic state of the GPIO pin input through the optocoupler. Then, using a module and a zero board for proof-of-concept (POC), it found that the relay works when the pull-up voltage is 5V. The table below shows the comparison of pull-up voltage value effects on the relay.

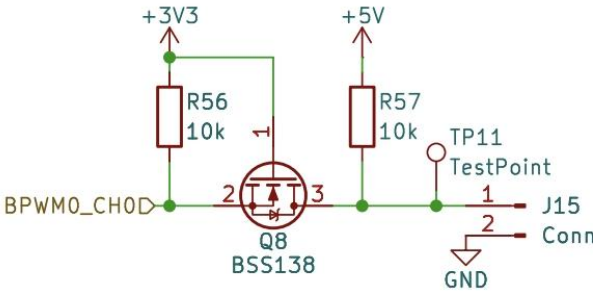
Pull-up voltage	Initial condition	GPIO HIGH	GPIO LOW
3V3	OFF	OFF	OFF
5V	OFF	ON	OFF
12V	ON	ON	ON



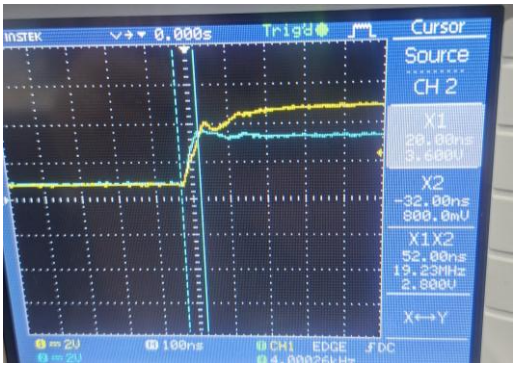
Corrected optoouopler footprint

In addition, it found that the optocoupler footprint is incorrect, where pins 3 and 4 were swapped. The image above shows the correct position of the pins. Reworking is required on the high-voltage side of the optocoupler and pull up voltage on the low-voltage side to make the relay functioning. After correcting the optocoupler wiring and replacing the pull-up voltage with 5V, the relay circuit is functioning as expected.

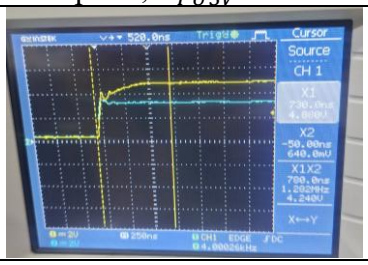
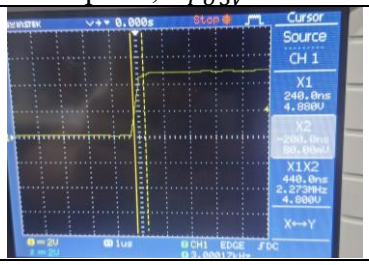
5.3. PWM



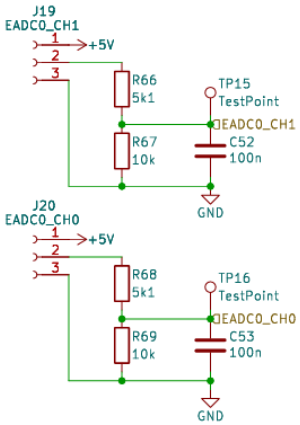
PWM test confirmed that the MCU was able to generate valid PWM signals on both BPWM0 and BPWM1. Measurements were first taken at the connector side to verify the final output path, and the waveforms were successfully observed, indicating that the PWM function was operating. Additional measurements were then performed at the GPIO side as a reference in case an issue occurred along the level-shifting stage. This comparison was useful to distinguish whether any waveform distortion originated from the MCU output itself or from the external interface circuit after the GPIO pin.



The results show that the PWM signal at the GPIO side had a very fast rising edge because it was driven directly by the MCU push-pull output. After passing through the BSS138-based logic level shifter, the rising edge became slower due to MOSFET parasitic capacitances and the RC charging effect at the 5 V node. The measured rise time remained small, but the effect was still observable on the oscilloscope. It was also found that the pull-up resistor on the 5 V side strongly influenced this behaviour: using a lower pull-up resistance produced a steeper edge and a shorter rise time, consistent with the RC time-constant relationship. Therefore, the PWM function itself can be considered operational, while the difference in edge shape is explained by the characteristics of the level-shifting circuit rather than by an error in PWM generation.

	Test point, $R_{PU5V} = 10k\Omega$	Test point, $R_{PU5V} = 1k\Omega$
Signal		
Rise time	700ns	440ns

5.4. ADC



The ADC test used a power supply to provide variable analogue values on the signal pin. Additional verification at the test point is performed to compare the digital values on the 5V and 3V3 scales, in case of an issue in the voltage divider circuit. The result is observed in a serial terminal.

5V Scale	3V3 Scale	ADC0	ADC1
0V	0V	0	0

1V	0.662V	865	867
2V	1.325V	1696	1699
2.5V	1.656V	2041	2043
3V	1.987V	2477	2482
4V	2.649V	3305	3308
5V	3.3V	4095	4095

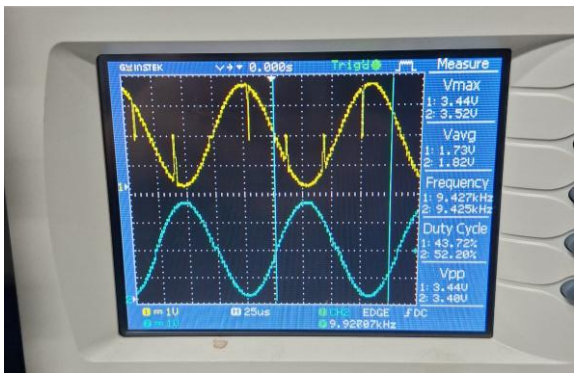
5.5. DAC

```
static const uint16_t g_au16Sine[] =
{
    2047, 2251, 2453, 2651, 2844, 3028, 3202, 3365, 3515, 3650, 3769, 3871, 3954,
    4019, 4064, 4088, 4095, 4076, 4040, 3984, 3908, 3813, 3701, 3573, 3429, 3272,
    3102, 2921, 2732, 2536, 2335, 2132, 1927, 1724, 1523, 1328, 1141,  962,  794,
     639,  497,  371,  262,  171,   99,   45,   12,    0,    7,   35,   84,  151,
     238,  343,  465,  602,  754,  919, 1095, 1281, 1475, 1674, 1876
};

static const uint32_t g_u32ArraySize = sizeof(g_au16Sine) / sizeof(uint16_t);
          /* Initial waveform data */
          DAC_WRITE_DATA(DAC0, 0, g_au16Sine[0]);
          DAC_WRITE_DATA(DAC1, 0, g_au16Sine[g_u32ArraySize / 2]);
```

The DAC test was performed using the group-mode sample program from the BSP, with the outputs observed at the corresponding DAC pins. The program uses a sine lookup table and interrupt-driven updates so that both DAC channels are converted simultaneously. In this configuration, DAC1 was intentionally initialized with a 180° phase shift relative to DAC0, which is consistent with the measured oscilloscope result showing two sinusoidal waveforms in opposite phase. The measured amplitudes and frequencies of both channels were also similar, indicating that the DAC peripheral and its basic output path were functioning properly.

A small spike was observed on the DAC1 waveform. This is likely caused by transient glitching during DAC code updates in the interrupt-driven group-mode output, rather than by a failure of the DAC itself. Since both channels still produced the expected sinusoidal signals with correct relative phase, the DAC test was considered successful. Based on this result, the tested PCB passed the basic DAC functionality verification.



5.6. UART

Independent UART testing was performed to verify the basic transmit functionality of the serial interfaces. UART0 was successfully tested using serial terminal output, where continuous counter data could be transmitted and displayed correctly, confirming that the UART initialization and basic TX path were functioning properly. In addition to this

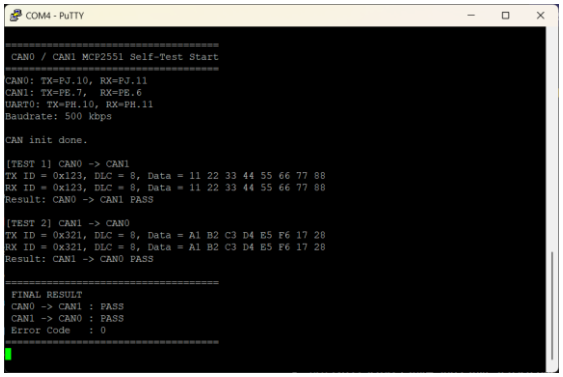
standalone test, UART0 was also used during ADC testing to monitor the digital conversion result, showing that it could support simple real-time data output for debugging purposes. UART2 was intended to be tested using the same method; however, it did not yet produce the expected result. This suggests that the remaining issue may be related either to the firmware configuration or to an incorrect pin order during the test setup, and therefore further verification is still required before concluding the UART2 hardware status.

5.7. CAN

```
static int32_t CAN_SendMsg(CANFD_T *psCanfd, uint32_t u32TxBuf, uint32_t u32Id, uint8_t *pu8Data,
{
    CANFD_FD_MSG_T sTxMsg;
    uint8_t i;

static int32_t CAN_WaitRx(CANFD_T *psCanfd, uint32_t u32RxBuf, uint32_t u32ExpectId, uint8_t *pu8
{
    CANFD_FD_MSG_T sRxMsg;
    uint32_t i;
        uint8_t au8Msg0[8] = {0x11, 0x22, 0x33, 0x44, 0x55, 0x66, 0x77, 0x88};
        uint8_t au8Msg1[8] = {0xA1, 0xB2, 0xC3, 0xD4, 0xE5, 0xF6, 0x17, 0x28};
        int32_t i32Ret;
```

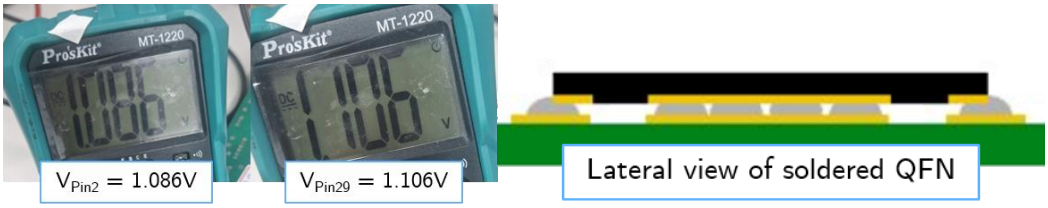
The CAN test was performed as a one-board communication test using two MCP2551 transceivers, where CAN0 transmitted data to CAN1 and the reverse path from CAN1 to CAN0 was also verified. The test firmware sent predefined data frames and checked whether the received ID, DLC, and payload matched the transmitted message. Based on the terminal output, both communication directions were completed successfully, indicating that the MCU CANFD peripheral, the external transceiver interface, and the basic CAN communication path were functioning correctly on the tested board.



The test result was monitored through UART0 using a serial terminal on the PC. In this test, the output was viewed in PuTTY, but any equivalent serial viewer could be used to display the UART messages. The terminal output showed the transmitted and received CAN data as well as the final pass status for both channels, confirming that the PCB passed the basic CAN communication test.

5.8. Ethernet

The Ethernet test was performed to verify the basic operation of the MCU EMAC interface, Ethernet PHY, RJ45 connection, and physical network link. The test firmware was based on the Nuvoton EMAC Tx/Rx BSP example. If DHCP is not available or fails due to the test environment, a static IP address can be used as an alternative for communication verification.



Before performing the ping test, several hardware conditions must be checked. The RJ45 port LEDs serve as the first indication of Ethernet link activity: the yellow LED indicates connectivity, and the green LED indicates data transmission. The PHY internal regulator outputs, including AVDD100OUT and DVDD100OUT, must also be checked because both should be around 1 V during normal operation. If these voltages are abnormal, the PHY may not establish a valid physical link. In addition, the exposed pad of the QFN PHY package must be properly soldered to the PCB ground pad, since it provides the main low-impedance ground return for the internal PHY regulators. If the exposed pad is floating, the internal regulator reference may become unstable and the Ethernet link may fail.

```
C:\Users\raiva>ping 192.168.137.50

Pinging 192.168.137.50 with 32 bytes of data:
Reply from 192.168.137.50: bytes=32 time=6ms TTL=64
Reply from 192.168.137.50: bytes=32 time=4ms TTL=64
Reply from 192.168.137.50: bytes=32 time=2ms TTL=64
Reply from 192.168.137.50: bytes=32 time=3ms TTL=64

Ping statistics for 192.168.137.50:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 2ms, Maximum = 6ms, Average = 3ms
```

The Ethernet communication is considered successful when the board obtains a valid IP address and responds to ping requests from the PC. If a link-local address such as 169.254.xxx.xxx appears, DHCP is considered failed because no valid DHCP server was assigned. If Internet Connection Sharing (ICS) is enabled on the PC, the board may receive an address with IP of 192.168.137.xxx. For field deployment, using a static IP address is preferred because it avoids dependence on DHCP availability and provides a predictable communication address.

6. Conclusion and Suggestion

6.1. Conclusion

This PCB version is conditionally functioning; reworking is required for relay circuit. However, it is not yet proper for field applications due to mechanical issues

6.2. Suggestion

- Conduct proof of concept (POC) before implementing a new circuit in a newer version
- Expand the buck module area to provide additional clearance for the module cables, if the module will be kept for the newer version
- Fix the relay circuit and its components' footprint
- Put all components on the top layer to ease the PCBA process

7. Revision History

NOTE: Page numbers for previous revisions may differ from page numbers in the current version.

Change made from Rev * (2026 April 27) to Rev A (2026 June 22)

- Updated relay test section (page 4)
- Updated ethernet test section (page 8)
- Updated conclusion (page 9)